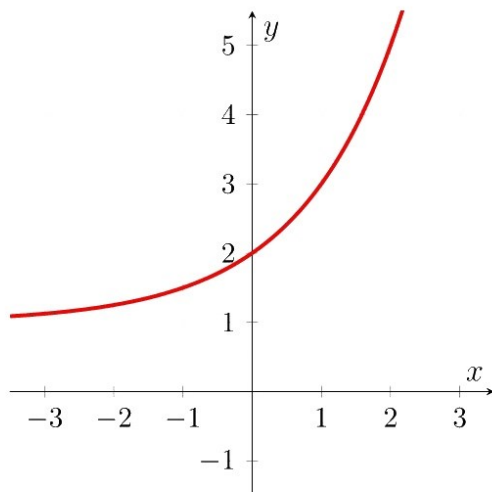


Exponential Functions

1. [March US 2023]



The graph of $y = f(x) - 1$ is shown. Which equation could define function f ?

- A) $f(x) = 2^x$
- B) $f(x) = 2^x - 1$
- C) $f(x) = 2^x + 1$
- D) $f(x) = 2^x + 2$

2. [June 2023]

An acceptable noise criterion rating for the background noise in a laundry room is 50. For a noise criterion rating of 50, the equation $y = 22(0.997)^{x-60} + 47$ gives the estimated sound pressure level, y , in decibels, as a function of the octave band center frequency, x , in hertz, where $x \geq 60$. Which of the following is the best interpretation of 47 in this context?

- A) 47 is 22 less than the estimated sound pressure level, in decibels, at an octave band center frequency of 60 hertz.
- B) 47 is 22 less than the estimated sound pressure level, in decibels, at an octave band center frequency of 0 hertz.
- C) 47 is the estimated sound pressure level, in decibels, at an octave band center frequency of 60 hertz.
- D) 47 is the estimated sound pressure level, in decibels, at an octave band center frequency of 0 hertz.

3. [March US 2023]

By examining pollen in the soil, scientists estimated that the number of *Ulmus* trees in an ancient population doubled every 664 years. There were n trees in the earliest known sample, where n is a constant. Which expression gives the estimated number of *Ulmus* trees x years after the year of the earliest known sample?

- A) $n(2)^{664x}$
- B) $n(2)^{\frac{664}{x}}$
- C) $n(2)^{\frac{x}{664}}$
- D) $n(2)^{(664+x)}$

4. [March US 2023]

$$f(x) = 5^x + 3$$

If the given function f is graphed in the xy -plane, where $y = f(x)$, what is the y -intercept of the graph?

- A) (0, 3)
- B) (0, 4)
- C) (0, 5)
- D) (0, 8)

5. [March US 2023]

Which table could represent values of x and their corresponding values of $f(x)$ for a decreasing exponential function f ?

A)

x	1	2	3	4
$f(x)$	1	2	3	4

B)

x	1	2	3	4
$f(x)$	8	4	2	1

C)

x	1	2	4	8
$f(x)$	1	9	17	25

D)

x	1	2	4	8
$f(x)$	25	17	9	1

6. [June 2023]

A certain neighborhood had a population of 1,340 in 2009. Each year for the next 5 years, the population of the neighborhood increased by approximately 3% of the population the previous year. Which of the following equations represents the population, N , of the neighborhood t years after 2009, where $t \leq 5$?

- A) $N = 0.03(1,340)^t$
- B) $N = 1,340(0.03)^t$
- C) $N = 1,340(1.03)^t$
- D) $N = 1.03(1,340)^t$

7. [August US 2023]

The function f is defined by $f(x) = a^x - b$, where a and b are constants. The graph of $y = f(x)$ in the xy -plane passes through the points $(0, -5)$ and $(1, 2)$. What is the value of a ?

- A) 8
- B) 7
- C) 6
- D) 5

8. [August US 2023]

$$f(x) = 6.4(0.5)^{3x}$$

A sample of carbon-11 radioactively decays. The function f models the mass, in grams(g), of carbon-11 in the sample x hours after it is initially measured. Which statement is the best interpretation of $f(2) = 0.1$?

- A) After 0.1 hour, the mass of carbon-11 in the sample is 2g.
- B) After 2 hour, the mass of carbon-11 in the sample is 0.1g.
- C) After 0.1 hour, the mass of carbon-11 in the sample has decreased by exactly 2g.
- D) After 2 hour, the mass of carbon-11 in the sample has decreased by exactly 0.1g.

9. [August US 2023]

Which table shows 3 values of x and their corresponding values of $f(x)$ for an increasing exponential function f ?

A)

x	$f(x)$
0	25
1	15
2	5

B)

x	$f(x)$
0	5
1	15
2	25

C)

x	$f(x)$
0	5
1	15
2	45

D)

x	$f(x)$
0	45
1	15
2	5

10. [August US 2023]

According to a model, the population of Canada's Yukon Territory was 33 thousand in 2008 and increased by 2% of the previous year's population for each of the next 4 years. Which function represents this model, where $f(x)$ is the population, in thousands, predicted by the model x years after 2008 and $0 \leq x \leq 4$?

- A) $f(x) = 33(x)^{1.02}$
- B) $f(x) = 33^{1.02x}$
- C) $f(x) = 33(0.02)^x$
- D) $f(x) = 33(1.02)^x$

11. [August US 2023]

$$f(x) = 7.3(0.98)^x$$

The function shown models the annual soil erosion rate, in tons per acre year, on US cropland x years since 1982. According to the model, which is the best interpretation of $7.3(0.98)^{10}$ in this context?

- A) The number of acres of US cropland in 1982
- B) The number of acres of US cropland in 1992
- C) The annual soil erosion rate, in tons per acre per year, in 1982
- D) The annual soil erosion rate, in tons per acre per year, in 1992

12. [December 2023]

The predicted number, $B(t)$, of a certain type of bacterial cell in a growth medium is given by the function $B(t) = 40(3)^t$, where t is the time, in days, since the number of bacterial cells was first measured. Which of the following is the best interpretation of $B(5) = 9,720$ in this context?

- A) 5 days after the number of bacterial cells was first measured, the predicted number of bacterial cells was 9,720
- B) 243 days after the number of bacterial cells was first measured, the predicted number of bacterial cells was 9,720
- C) 9,720 days after the number of bacterial cells was first measured, the predicted number of bacterial cells was 5
- D) 9,720 days after the number of bacterial cells was first measured, the predicted number of bacterial cells was 243

13. [November 2023]

A model estimates that the initial number of photos in a beam is 800 when the beam reaches the surface of a certain material. The model also estimates that when the beam passes through this material the number of photos in the beam decreases by 50% for each 13 millimeters of the material the beam passes through. Which equation represents this model, where I is the estimated number of photos in the beam when the beam reaches a point x millimeters from the surface?

- A) $I = 800(0.5)^{\frac{x}{13}}$
- B) $I = 800(0.5)^x$
- C) $I = 800(2)^{\frac{x}{13}}$
- D) $I = 800(13)^{\frac{x}{2}}$

14. [December 2023]

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(2,0)$ and y -intercept at $(0,-255)$. What is the value of b ?

15. [December 2023]

$$f(t) = 7,000(0.67)^t$$

The given function f models the number of catalogs a company sent to their customers at the end of each year, where t represents the number of years since the end of 1995, and $0 \leq t \leq 5$. If $y = f(x)$ is graphed in the ty -plane, which of the following is the best interpretation of the y -intercept of the graph in this context?

- A) The minimum estimated number of catalogs the company sent to their customers during 5 year is 1,411
- B) The minimum estimated number of catalogs the company sent to their customers during the 5 years was 7,000
- C) The estimated number of catalogs the company sent to their customers at the end of 1995 was 1,411
- D) The estimated number of catalogs the company sent to their customers at the end of 1995 was 7,000

16. [November 2023]

A savings account is opened with an initial deposit of \$6,000. The amount of money in the account t years after the initial deposit is given by the function $f(t) = 6,000(1.01)^{2t}$. Which of the following is the best interpretation of the statement " $f(5)$ is approximately equal to 6,627.73" in this context?

- A) 5 years after the initial deposit, the amount of money, in dollars, in the account is 6,627.73
- B) Every 5 years, the amount of money, in dollars, in the account increases by 6,627.73
- C) 5 years after the initial deposit, the amount of money, in dollars, in the account has increased by 6,627.73
- D) Every 5 years, the amount of money, in dollars, in the account decreases by 6,627.73

17. [December 2023]

A certain neighborhood had a population of 1,130 in 2006. Each year for the next 5 years the population of the neighborhood increased by approximately 4% of the population of the previous year. Which of the following equations represents the population N , of the neighborhood t years after 2006, where $t \leq 5$?

- A) $N = 0.04(1,130)^t$
- B) $N = 1,130(0.04)^t$
- C) $N = 1,130(1.04)^t$
- D) $N = 1.04(1,130)^t$

18. [March 2024]

$$f(x) = 6,000(0.70)^x$$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 6,000 in this context?

- A) The estimated percent decrease in the insect population for this species and area every 8 years after 2008
- B) The estimated percent decrease in the insect population for this species and area each year after 2008
- C) The estimated insect population for this species and area 8 years after 2008
- D) The estimated initial insect population for this species and area in 2008

19. [March 2024]

The function f is defined by $f(x) = 42(0.24)^x$. For any positive integer n , the value of $f(n)$ is $p\%$ less than the value of $f(n-1)$. What is the value of p ?

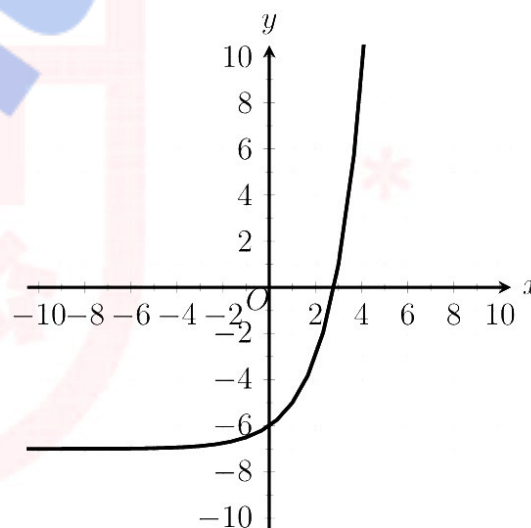
- A) 76
- B) 58
- C) 42
- D) 24

20. [March 2024]

In a certain state, the population of pheasants, a type of bird, is estimated each year by counting the number of pheasants observed along certain roads in the state. On average, each year from 2005 to 2015 the number of pheasants counted per mile of road decreased by 3.5% of the number of pheasants per mile of road the previous year. Based on this average, if there were 6.97 pheasants per mile of road in this state in 2005, which of the following best approximates the number of pheasants per mile of road in 2015?

- A) $0.965(6.97)^{10}$
- B) $1.035(6.97)^{10}$
- C) $6.97(0.035)^{10}$
- D) $6.97(0.965)^{10}$

21. [March 2024]



The graph of the equation $y = 2^x + k$ is shown, where k is a constant. What is the value of k ?

- A) -7
- B) -6
- A) 7
- B) 6

22. [May US 2024]

What is the y -intercept of $y = -(7)^x - 22$ in the xy -plane?

- A) (0,-23)
- B) (0,-29)
- C) (0,-22)
- D) (0,-21)

23. [March 2024]

At the time of posting a video, a social media channel had 95 subscribers. Each day for five days after the video was posted, the number of subscribers doubled from the previous day. Which equation gives the total number of subscribers, n , to the channel d days after the video was posted?

- A) $n = (95)^d$ C) $n = 95 \left(\frac{1}{2}\right)^d$
 B) $n = 95(2)^d$ D) $n = (95)^2 + d$

24. [March 2024]

$$q(x) = 44(4^x)$$

Which table gives three values of x and their corresponding values of $q(x)$ for function q ?

A)

x	-1	0	1
$q(x)$	176	0	176

B)

x	-1	0	1
$q(x)$	$\frac{1}{11}$	4	176

C)

x	-1	0	1
$q(x)$	$\frac{1}{11}$	44	176

D)

x	-1	0	1
$q(x)$	11	44	176

25. [March 2024]

At the time of posting a video, a social media channel ha 88 subscribers. Each day after the video was posted, the number of subscribers doubled from the number of subscribers on the previous day. Which equation gives the total number of subscribers, n , to the channel d days after the videos was posted?

- A) $n = (88)^d$
 B) $n = 88(2)^d$
 C) $n = 88\left(\frac{1}{2}\right)^d$
 D) $n = 88^2 + d$

26. [March 2024]

The estimated number of a certain bacteria $B(t)$ in a petri dish is given by the equation $B(t) = 6700(2)^t$, where t is the time, in hours, after the bacteria were added to the petri dish. Which statement is the best interpretations of $B(4) = 107200$ in this context?

- A) 0 hours after the bacteria were added to the petri dish, the estimated number of bacteria in the petri dish is 4.
 B) 0 hours after the bacteria were added to the petri dish, the estimated number of bacteria in the petri dish is 107200.
 C) 4 hours after the bacteria were added to the petri dish, the estimated number of bacteria in the petri dish is 4.
 D) 4 hours after the bacteria were added to the petri dish, the estimated number of bacteria in the petri dish is 107200.

27. [March 2024]

$$q(x) = 12(3)^x$$

Which table gives three values of x and their corresponding values of $q(x)$ for function q ?

A)

x	-1	0	1
$q(x)$	4	12	36

B)

x	-1	0	1
$q(x)$	$\frac{1}{4}$	12	36

C)

x	-1	0	1
$q(x)$	$\frac{1}{4}$	3	36

D)

x	-1	0	1
$q(x)$	-36	0	36

28. [May 2024]

The function h is defined by $h(x) = a^x + b$, where a and b are positive constants. The graph of $y = h(x)$ in the xy -plane passes through the points $(0, 8)$ and $(2, 32)$. What is the value of ab ?

- A) 32 C) 40
 B) 35 D) 64

29. [May 2024]

$$f(x) = k(1.84)^x$$

The function f is defined by the given equation, where k is a constant. The value of $f(x)$ increases by $p\%$ for every increase of x in 1. For which of the following functions, where k is a constant, does the value of $g(x)$ increase by $p\%$ for every increase of x in 4?

- A) $g(x) = k(1.84^x)^{\frac{1}{4}}$
- B) $g(x) = k(1.84^x)^4$
- C) $g(x) = k(1.84)^{x-4}$
- D) $g(x) = k(1.84)^{x+4}$

30. [May 2024]

For a certain isotope, at the end of every 29-year period, the mass of a sample of the isotope has decreased to half its mass at the beginning of the 29-year period. The function $s(t) = 216(0.5)^{\frac{t}{29}}$ gives the approximate mass of this isotope, in grams, remaining t years after a 216-gram sample starts to decay. Which statement is the best interpretation of $s(3 \cdot 29) = 27$ in this context?

- A) The mass of the sample is approximately 27 grams 3 years after the sample starts to decay.
- B) The mass of the sample is approximately 27 grams 3 29-year periods after the sample starts to decay.
- C) The mass of the sample is approximately 3 grams 27 29-year periods after the sample starts to decay.
- D) The mass of the sample is approximately 3 grams 27 years before the sample starts to decay.

31. [May US 2024]

In a certain state, the population of pheasants, a type of bird, is estimated each year by counting the number of pheasants observed along certain roads in the state. On average, each year from 2005 to 2015 the number of pheasants counter per mile of road decreased by 3.5% of the number of

pheasants per mile of road the previous year. Based on this average, if there were 6.93 pheasants per mile of road in this state in 2005, which of the following best approximates the number of pheasants per mile of road in 2015?

- A) $6.93(0.965)^{10}$
- B) $6.93(0.035)^{10}$
- C) $1.035(6.93)^{10}$
- D) $0.965(6.93)^{10}$

32. [May US 2024]

At the start of an experiment, the pressure inside a container was 19 pounds per square inch (psi). The pressure inside the container doubled every 28 minutes after the start of the experiment. Which equation represents this situation, where y is the pressure, in psi, inside the container x minutes after the start of the experiment?

- A) $y = 19(2)^{\frac{x}{28}}$
- B) $y = 19(28)^{2x}$
- C) $y = 19(x + 28)^2$
- D) $y = 19x^2 + 28$

33. [May US 2024]

x	$h(x)$
0	1.46
2	2.32
4	3.68

The table shows exponential relationship between the number of years, x , since Isela started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

- A) $h(x) = 1.26(0.46)^x$
- B) $h(x) = 1.26(1.46)^x$
- C) $h(x) = 1.46(0.26)^x$
- D) $h(x) = 1.46(1.26)^x$

34. [May 2024]

x	$h(x)$
0	1.25
2	1.80
4	2.59

The table shows the exponential relationship between the number of years, x , since Hana started training in pole vault, and the estimated height $h(x)$, in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

- A) $h(x) = 1.20(0.25)^x$
- B) $h(x) = 1.20(1.25)^x$
- C) $h(x) = 1.25(0.20)^x$
- D) $h(x) = 1.25(1.20)^x$

35. [May US 2024]

$$f(x) = 2,000(0.85)^x$$

A conservation scientist implemented a program to reduce a certain invasive insect population in an area. The given function estimates this insect species' population x years after 2008, where $x \leq 8$. Which of the following is the best interpretation of 2,000 in this context?

- A) The estimated percent decrease in the insect population for this species and area every 8 years after 2008
- B) The estimated percent decrease in the insect population for this species and area each year after 2008
- C) The estimated insect population for this species and area 8 years after 2008
- D) The estimated initial insect population for this species and area in 2008

36. [June 2024]

The function $f(t) = 10,000(2)^{\frac{t}{330}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

37. [May 2024]

The function $P(x) = 533(1.011)^x$ models the population of a certain town each year from 1951 through 1966, where x is the number of years after 1951. Which of the following is the best interpretation of " $P(5)$ is approximately equal to 563" in this context?

- A) The population of the town is estimated to be approximately 5 greater in 1956 than in 1951.
- B) The population of the town is estimated to be approximately 563 in 1956.
- C) The population of the town is estimated to be approximately 5 times greater in 1956 than in 1951.
- D) The population of the town is estimated to increase by approximately 563 every 5 years between 1951 and 1956.

38. [May 2024]

$$P(t) = 260(1.03)^{\left(\frac{6}{n}\right)t}$$

The function P models the population, in thousands, of a certain city t years after 2006. According to the model, the population is predicted to increase by 3% every n months. What is the value of n ?

- A) 8
- B) 12
- C) 18
- D) 72

39. [May 2024]

The number of bacteria in a liquid medium doubles every day. There are 72,000 bacteria in the liquid medium at the start of an observation. Which of the following represents the number of bacteria, y , in the liquid medium t days after the start of the observation?

- A) $y = \frac{1}{2}(72,000)^t$
- B) $y = 2(72,000)^t$
- C) $y = 72,000\left(\frac{1}{2}\right)^t$
- D) $y = 72,000(2)^t$

40. [June 2024]

A company has a customer loyalty program. In January 2018, there were 900 customers enrolled in the loyalty program. For the next 24 months after January 2018, the total number of customers enrolled in the loyalty program each month was 2% greater than the total number enrolled the previous month. Which equation gives the total number of customers, c , enrolled in the company's loyalty program m month after January 2018, where $m \leq 24$?

- A) $c = 900(0.02)^m$
- B) $c = 900(1.02)^m$
- C) $c = 900(1.2)^m$
- D) $c = 900(2)^m$

41. [June 2024]

A company has a customer loyalty program. In January 2018, there were 1,600 customers enrolled in the loyalty program. For the next 24 months after January 2018, the total number of customers enrolled in the loyalty program each month was 5% greater than the total number enrolled the previous month. Which equation gives the total number of customers, c , enrolled in the company's loyalty program m months after January 2018, where $m \leq 24$?

- (A) $c = 1,600(0.05)^m$
- (B) $c = 1,600(1.05)^m$
- (C) $c = 1,600(1.5)^m$
- (D) $c = 1,600(5)^m$

42. [June 2024]

$$f(x) = 24(1.30)^{\frac{x}{3}}$$

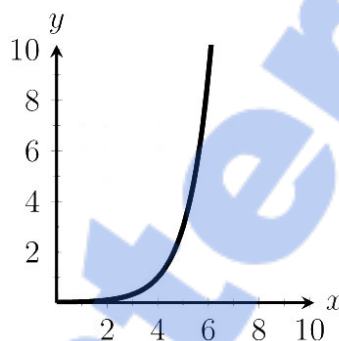
For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p ?

- A) 30
- B) 41
- C) 60
- D) 69

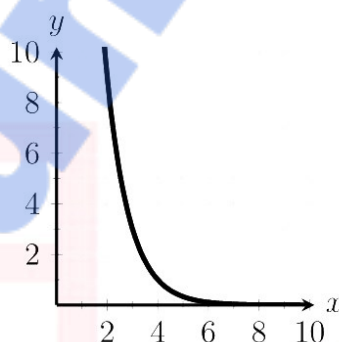
43. [June 2024]

Which of the following is a graph of a decreasing exponential function?

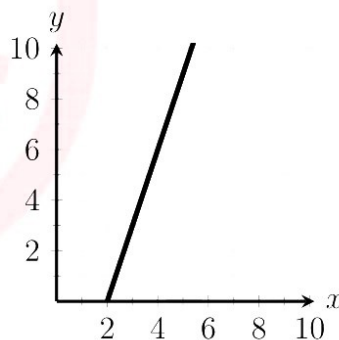
A)



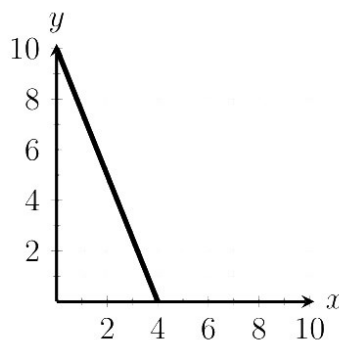
B)



C)



D)



44. [June 2024]

The function g is defined by $g(x) = 10^{3x-1}$. What is the value of x when $g(x)$ is equal to 10,000?

- A) $\frac{4}{3}$ C) 3
B) $\frac{5}{3}$ D) 4

45. [June 2024]

The function f is defined by

$$f(x) = a(3.1^x + 3.1^b),$$

where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f , where k and m are constants. Which of the following equations displays the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane as a constant or coefficient?

- I. $g(x) = a(3.1^x + k)$
II. $h(x) = a(3.1)^x + m$
A) I only
B) II only
C) I and II
D) Neither I nor II

46. [August 2024]

The y -intercept of the graph of $y = 81 + 65^x$ in the xy -plane is $(0, y)$. What is the value of y ?

47. [August US 2024]

For the exponential function f , the value of $f(1)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

- A) $f(x) = 50(1.4)^{x+1}$
B) $f(x) = 70(1.4)^x$
C) $f(x) = 98(1.4)^{x-1}$
D) $f(x) = 137.2(1.4)^{x-2}$

48. [June 2024]

The function $f(t) = 10,000(2)^{\frac{t}{50}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

49. [August 2024]

The number of bacteria in a liquid medium doubles every day. There are 72,000 bacteria in the liquid medium at the start of an observation. Which of the following represents the number of bacteria, y , in the liquid medium t days after the start of the observation?

- A) $y = \frac{1}{2}(72,000)^t$
B) $y = 2(72,000)^t$
C) $y = 72,000(\frac{1}{2})^t$
D) $y = 72,000(2)^t$

50. [August 2024]

According to a model, the estimated volume of a certain Douglas fir tree increased by 15.5% for every increase of 100 centimeters in the tree's height. The equation

$$y = ab^{\frac{t}{100}}$$

represents this model, where y represents the estimated volume, in cubic centimeters, of this Douglas fir tree, x represents the tree's height, in centimeters, and a and b are constants. In this equation, what is the value of b ?

- A) 0.155 C) 1.155
B) 1 D) 15.5

51. [August US 2024]

$$f(x) = 7020(0.28)^{\frac{x}{12}}$$

The function f gives the value of a certain piece of equipment after x months of use. If the value of the equipment decreases each year by $p\%$ of its value the preceding year, what is the value of p ?

- A) 2 C) 28
B) 10 D) 72

52. [August 2024]

The functions f and g are defined by the given equations, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the maximum value of the function it defines, where $x \geq 0$?

1. $f(x) = 16(1.24)^x + 43$

2. $g(x) = 8(0.72)^x$

- A) 1 only
 B) 2 only
 C) 1 and 2
 D) Neither 1 nor 2

53. [August 2024]

For the function h , the table gives three values of x and their corresponding values of $h(x)$. Which equation could define h ?

x	$h(x)$
0	30
1	31
2	33

- A) $y = 2^x$
 B) $y = 2^x + 2$
 C) $y = 2^x + 28$
 D) $y = 2^x + 29$

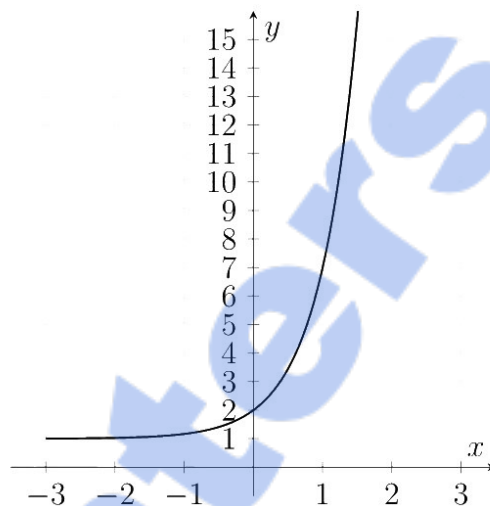
54. [August 2024]

$$f(x) = 240(2)^x$$

The function f is defined by the given equation. If $g(x) = f(x - 3)$, which of the following equations defines the function g ?

- A) $g(x) = 30(2)^x$
 B) $g(x) = 80(2)^x$
 C) $g(x) = 240(8)^x$
 D) $g(x) = 240(6)^x$

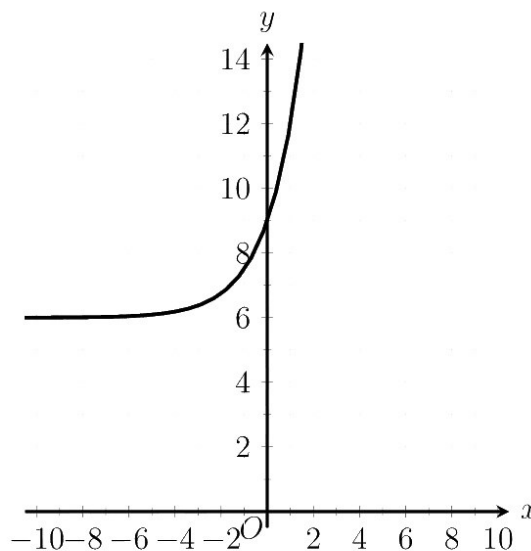
55. [August 2024]



The graph of $y = f(x) + 1$ is shown. Which equation defines the function f ?

- A) $f(x) = 6^x$
 B) $f(x) = 2^x$
 C) $f(x) = 6^x + 2$
 D) $f(x) = 2^x + 1$

56. [October US 2024]



What is the y-intercept of the graph shown?

- A) $(-9, 0)$
 B) $(-6, 0)$
 C) $(0, 6)$
 D) $(0, 9)$

57. [August US 2024]

Time(years)	Total amount (dollars)
0	867.00
1	870.47
2	873.95

Sam opened a savings account at a bank. The table shows the exponential relationship between the time t , in years, since Sam opened the account and the total amount n , in dollars, in the account. If Sam made no additional deposits or withdrawals, which of the following equations best represents the relationship between t and n ?

- A) $n = 0.004(1 + 867)^t$
- B) $n = 867(1 + 0.004)^t$
- C) $n = (1 + 0.004)^t$
- D) $n = (1 + 867)^t$

58. [October US 2024]

The function $f(x) = 242(1.040)^x$ models the value, in dollars, of a certain bank account from 1953 through 1968, where x is the number of years after the end of 1953. Which of the following is the best interpretation of $f(5)$, which is approximately equal to 294, in this context?

- A) The value of the bank account is estimated to be approximately 5 dollars greater at the end of 1958 than at the end of 1953.
- B) The value of the bank account is estimated to be approximately 294 dollars at the end of 1958.
- C) The value, in dollars, of the bank account is estimated to be approximately 5 times greater at the end of 1958 than at the end of 1953.
- D) The value of the bank account is estimated to increase by approximately 294 dollars every 5 years between the end of 1953 and the end of 1968.

59. [August US 2024]

The number of bacteria in a liquid medium doubles every hour. There are 44000 bacteria in the liquid medium at the start of the observation. Which of the following represents number of bacteria, y , in the liquid medium t hours after the the start of the observation?

- A) $y = \frac{1}{2}(44000)^t$
- B) $y = 2(44000)^t$
- C) $y = 44000(\frac{1}{2})^t$
- D) $y = 44000(2)^t$

60. [August US 2024]

The function f and g are defined by the equations, where $x > 0$. Which of the following equations displays, as a constant or a coefficient, the maximum value of the function it defines, where $x \geq 0$?

- I. $f(x) = 16(1.25)^x + 43$
- II. $g(x) = 6(0.75)^x$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

61. [August US 2024]

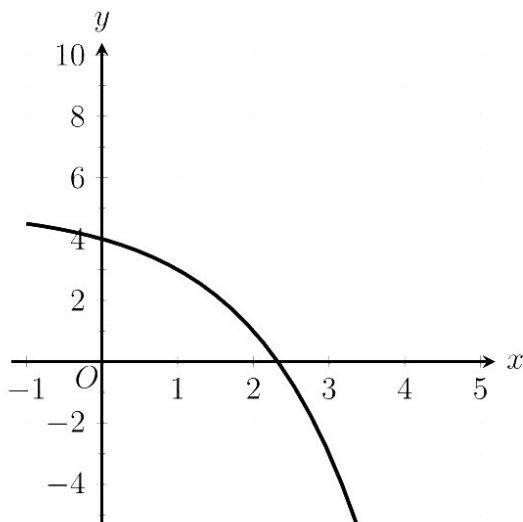
The function

$$f(t) = 80,000(2)^{\frac{t}{790}}$$

gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

- A) 2
- B) 790
- C) 1,580
- D) 80,000

62. [October US 2024]



The graph of $y = f(x) + 2$ is shown. Which equation defines function f ?

- A) $f(x) = -2^x + 1$
- B) $f(x) = -2^x + 3$
- C) $f(x) = -2^x + 4$
- D) $f(x) = -2^x + 5$

63. [October US 2024]

The functions f and g are defined by the equations shown, where a and b are integer constants, $a < b$ and $b < 0$. If $y = f(x)$ and $y = g(x)$ are graphed in the xy -plane, which of the following equations displays, as a constant or coefficient, the y -coordinate of the y -intercept of the graph of the corresponding function?

I. $f(x) = a(3.3)^{x+b}$

II. $g(x) = a(3.3)^x + b$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

64. [October 2024]

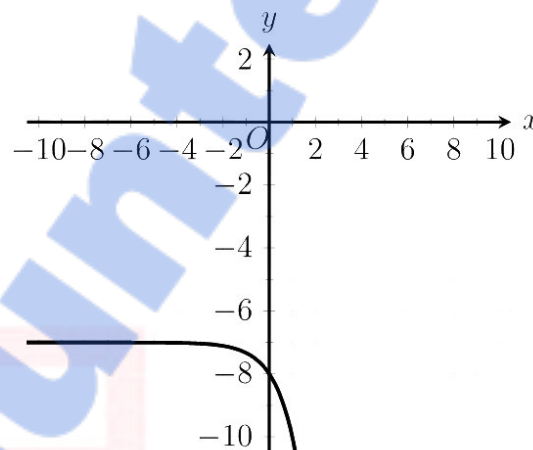
The functions f and g are defined by the given equations below, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $f(x) = 18(1.4)^{x+5}$

II. $g(x) = 18(1.16)(1.4)^{x+3}$

- A) I only
- B) II only
- C) I and II
- D) Neither I nor II

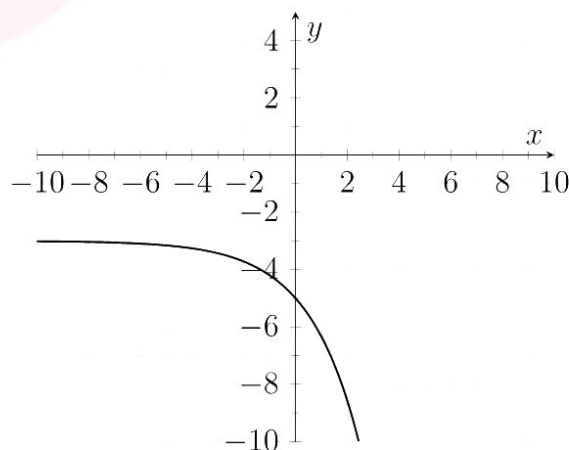
65. [October 2024]



The graph of $y = f(x)$ is shown, where $f(x) = ab^x + c$, and a , b , and c are constants. For how many values of x does $f(x) = 0$?

- A) Three
- B) Two
- C) One
- D) Zero

66. [October US 2024]



The graph of $y = f(x) = ab^x + c$ is shown, and a , b and c are constants. For how many value of x does $f(x) = 0$?

- A) Three
- B) Two
- C) One
- D) Zero

67. [October US 2024]

In a certain state, the population of pheasants, a type of bird, is estimated each year by counting the number of pheasants observed along certain roads in the state. On average, each year, from 2005 to 2015 the number of pheasants counted per mile of road decreased by 3.5% of the number of pheasants per mile of the road the previous year. Based on this average, if there were 6.32 pheasants per mile of road in this state in 2005, which of the following best approximates the number of pheasants per mile in 2015?

- A) $0.956(6.32)^{10}$
- B) $1.035(6.32)^{10}$
- C) $6.32(0.035)^{10}$
- D) $0.632(0.965)^{10}$

68. [November US 2024]

For an online trivia game, 490 points are awarded for a correct answer if a question is answered in less than 5 seconds from the time the question is asked. Each 5 seconds after the question is asked, the number of points awarded for a correct answer decreases by 20% of the number of points awarded for a correct answer in the previous 5 seconds. Which function gives the number of points awarded for a correct answer x seconds from the time the question is asked, where x is a multiple of 5?

- A) $f(x) = 490(0.80)^{\frac{x}{5}}$
- B) $f(x) = 490(0.80)^{5x}$
- C) $f(x) = 490(0.20)^{\frac{x}{5}}$
- D) $f(x) = 490(0.20)^{5x}$

69. [November US 2024]

What is the y -intercept of the graph of $y = 12(5)^{x+2}$ in the xy -plane?

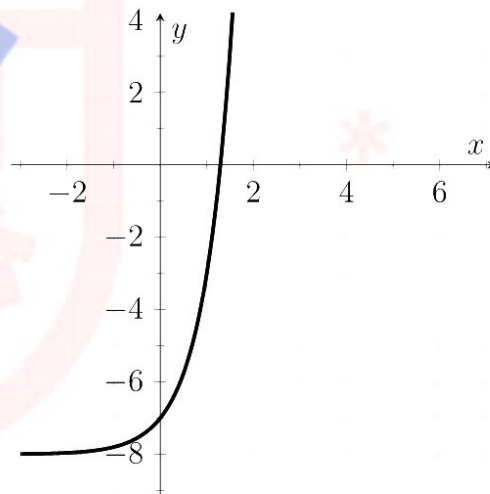
- A) (0, 12)
- B) (0, 120)
- C) (0, 300)
- D) (0, 3,600)

70. [November 2024]

In a certain state the population of pheasants, a type of bird, is measured by counting the number pheasants observed along certain roads in the state. On average, each year from 2005 to 2015 the number of pheasants counted per mile decreased by 3.5% of the number of pheasants per mile of road the previous year. Based on this average, if there were 6.22 pheasants per mile of road in this state in 2005, which of the following best approximates the number of pheasants per mile of road in 2015?

- A) $0.965(6.22)^{10}$
- B) $1.035(6.22)^{10}$
- C) $6.22(0.035)^{10}$
- D) $6.22(0.965)^{10}$

71. [October US 2024]



The graph of the equation $y = 5^x + k$ is shown, where k is a constant. What is the value of k ?

- A) -8
- B) -7
- C) 7
- D) 8

72. [October US 2024]

$$y = 7900(0.86)^x$$

The given equation gives the value, in dollars, of a certain car x years after it was purchased. According to the given equation, what is the estimated value, in dollars, of the car at the time it was purchased?

- A) 790 C) 7900
B) 6794 D) 8600

73. [October US 2024]

A model predicts that the population of a certain town was 3,000 in 2007. The model also predicts that each year for the next 5 years, the population p increased by 8% of the previous year's population. Which equation best represents this model, where t is the number of years after 2007, for $0 \leq t \leq 5$?

- (A) $p = 0.92(3,000)$
(B) $p = 1.08(3,000)$
(C) $p = 3,000(0.92)^t$
(D) $p = 3,000(1.08)^t$

74. [November US 2024]

In the xy -plane, the graph of the equation $y = 9^x$ is translated 8 units down. What is an equation of the resulting graph?

- A) $y = 9^x + 8$ C) $y = 9^{(x+8)}$
B) $y = 9^x - 8$ D) $y = 9^{(x-8)}$

75. [November 2024]

A baker containing a liquid is placed on a table. The function

$$g(t) = 294 + (363 - 294)(2.72)^{-0.103t}$$

gives the approximate temperature, in kelvins, of the liquid t minutes after the baker was placed on the table. According to this function, what was the approximate temperature, in kelvins, of the liquid when the baker was placed on the table?

76. [December US 2024]

The population of a certain city doubled every 25 years from 1660 to 1760. The population of this city was 208,000 in 1760. What was the population this city in 1660?

- A) 13,000 C) 104,000
B) 26,000 D) 416,000

77. [November 2024]

The function f is defined by

$$f(x) = a^x + b,$$

where a and b are constants and $a > 0$. In the xy -plane, the graph of $y = f(x)$ has a y -intercept at $(0, -20)$ and passes through the point $(2, 43)$. What is the value of $a - b$?

78. [December US 2024]

The function

$$f(x) = 600(0.5)^{\frac{x}{11}}$$

gives the predicted intensity of a beam $f(x)$, in number of photons in the beam, x millimeters below the surface of a certain material. What is the best interpretation of $f(11) = 300$?

- A) A beam at the surface of the material has a predicted intensity of 11 photons in the beam.
B) A beam at the surface of the material has a predicted intensity of 300 photons in the beam.
C) A beam 11 millimeters below the surface of the material has a predicted intensity of 300 photons in the beam.
D) A beam 300 millimeters below the surface of the material has a predicted intensity of 600 photons in the beam.

79. [December US 2024]

The function h is defined by

$$h(x) = 14(2)^{-x} + 7.$$

What is the value of $h(3)$?

- A) $\frac{42}{4}$ C) -77
B) $\frac{35}{4}$ D) -105

80. [December US 2024]

What is the y -intercept of the graph of $y = 16^{x+3}$ in the xy -plane?

- A) (0, 4,096) C) (0, 16)
B) (0, 0) D) (0, 3)

81. [December US 2024]

$$f(x) = 25(1.40)^{\frac{x}{3}}$$

For the given function f , the value of $f(x)$ increases by $p\%$ for every increase of x by 6. What is the value of p ?

- A) 40 C) 80
B) 49 D) 96

82. [December US 2024]

A computer program models the population of a certain insect in an environment where the insect has no natural predators. According to the model, the estimated total mass of the population of insects at the end of every 5-week period is 158% greater than the estimated total mass of the population of insects at the end of the previous 5-week period. The estimated total mass of the population of insects at the end of 15 weeks is 627.7 grams.

Which equation best represents this model, where M is the estimated total mass, in grams, of the population of insects at the end of t weeks?

- A) $M = 538.92(1.58)^{\frac{5}{t}}$
B) $M = 457.66(2.58)^{\frac{5}{t}}$
C) $M = 159.14(1.58)^{\frac{t}{5}}$
D) $M = 36.55(2.58)^{\frac{t}{5}}$

83. [December 2024]

The population of a certain city doubled every 75 years from 1902 to 1952. The population of this city was 160,000 in 1952. What was the population of this city in 1902?

84. [December 2024]

The relationship between the variables x and y is defined by an exponential equation. When $x = 0$, the value of y is 90, and for every increase in the value of x by 1, the corresponding value of y increases by 20% of its previous value. Which equation represents this relationship?

- A) $y = 90(1.20)^x$
B) $y = 90(1.02)^x$
C) $y = 20(1.90)^x$
D) $y = 20(1.09)^x$

85. [December 2024]

The equation $y = 500(2)^{2x}$ gives the estimated number of online newsletter subscribers at the end of every six-month period, x , starting from the end of January 1997, where $0 \leq x \leq 4$. Which statement is the best interpretation of the y -intercept of the graph of this equation in the xy -plane?

- A) The estimated number of online newsletter subscribers at the end of the first six-month period was 500.
B) The estimated number of online newsletter subscribers at the end of January 1997 was 500.
C) The estimated number of online newsletter subscribers at the end of January 1998 was 1,000.
D) The estimated number of online newsletter subscribers at the end of January 1997 was 1,000.

86. [December 2024]

$$g(x) = \frac{1}{9}(5)^x$$

If the given function g is graphed in the xy -plane, where $y = g(x)$, what is the y -intercept of the graph?

- A) (0,0) C) $(0, \frac{5}{9})$
B) $(0, \frac{1}{9})$ D) (0,5)

87. [December 2024]

$$z(w) = (0.829)^{2w}$$

The function z is defined by the equation shown. The value of $z(w)$ decreases by $p\%$ for each increase by 1 in the value of w . Which of the following is closest to the value of p ?

- A) 0.171 C) 17.1
B) 0.829 D) 31.3

88. [December 2024]

$$f(x) = 66(b)^x$$

In the given function, b is a positive constant. For every increase in the value of x by 1, the value of $f(x)$ increases by $c\%$, where $0 < c < 100$. Which expression gives the value of c in terms of b ?

- A) $1 + \frac{b}{100}$ C) $100(b + 1)$
B) $b + 100$ D) $100(b - 1)$

89. [December 2024]

The function d is defined by

$$d(x) = 200 - 6^x.$$

What is the value of $d(0)$

- A) 140 C) 199
B) 194 D) 200

90. [December 2024]

The function f is defined by

$$f(x) = (-7)(4)^x + 31.$$

What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- A) (0,24) C) (0,31)
B) (0,4) D) (0,-7)

91. [December 2024]

The equation $N(m) = 65(Q)^{\frac{m}{4}}$ gives the predicted population $N(m)$, in thousands, of a certain bacteria colony m minutes after the initial measurement, where Q is a constant greater than 1. The predicted population increases by $p\%$ every 120 seconds. What is the value of p in terms of Q ?

- A) $100(Q^{\frac{1}{2}} + 1)$ C) $100(Q^{\frac{1}{2}} - 1)$
B) $100(Q^{30} + 1)$ D) $100(Q^{30} - 1)$